

Project Proposal Form

Institute of Information Technology (IIT)

University of Dhaka

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MIT Open Credit Management System

Project Description

The MIT Open Credit Management System is a web platform designed to automate the student lifecycle for IIT, Dhaka. It replaces manual paperwork with secure dashboards to manage course enrollments, enforce seat limits, calculate penalty fees, process midterm and final marks and safely store academic records.

Problem Statement

Manually tracking student credit limits, course rules, and fee payments causes errors and delays. When financial and academic data aren't synced, students face frustrating registration blocks.

Furthermore, manual calculation of subject marks and CGPA across multiple semesters often leads to grading inaccuracies.

This project solves these issues by creating a centralized system that handles all checks and updates and result processing automatically in real-time.

Objectives

- 12-Credit Rule:** Automatically ensure students take exactly 12 credits per semester (36 total).
- 40-Seat Limit:** Strictly enforce a 40-seat maximum for optional courses.
- Automated Fees:** Automatically calculate fees for retakes and semester gaps.
- Grading & CGPA Module:** Provide a tabular interface for teachers to input midterm and final marks (by Registration and Roll number) to automatically generate the subject grade and overall CGPA
- Clear Dashboards:** Provide simple screens for students and admins to track progress, and results.
- Secure & Simple:** Ensure secure logins and easy deployment.

Proposed Solution

- We will build a reliable, easy-to-use web application. The frontend will allow students to easily register for courses and check their dues, and view their final results. For the faculty and admins, the system will offer a secure tabular data-entry interface to easily input student marks. Behind the scenes, the backend will automatically verify all academic rules (like credit limits and seat availability) and instantly calculate the correct CGPA before confirming any action.

Key Technologies & Tools

Frontend: Angular TypeScript

Backend: Spring Boot

Database: PostgreSQL.

Supervisor's Name: Dr. Md. Nurul Ahad Tawhid

Signature of the supervisor:

Date: 19th May, 2026

Proposal Presentation Feedback:

Midterm Presentation Feedback:

